

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVC603 - TRANSPORTATION ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1) Which type of bridge foundation is suitable when rock level close to the ground or top soil strata is strong ? a) Deep foundations b) Shallow foundations c) Pile foundation d) Well foundation
- 2) Consider the following statements, Excessive camber is not provided on the roads because 1.transverse tire causes discomfort 2.of formation of cross ruts 3.of likely toppling over of highly laden bullock carts 4.of higher cost involved Which of these statements are correct ? a) 2, 3 and 4 b) 1, 3 and 4 c) 1, 2 and 4 d) 1, 2 and 3
- 3) Ruling gradient on highways as per IRC in plain terrain is a) 1 in 30 b) 1 in 60 c) 1 in 100 d) 1 in 200
- 4) If the difference in elevation between the edges of a pavement of width 9 m and its crown is 7.5 cm, what is the camber of the pavement ? a) 1 in 60 b) 1 in 45 c) 1 in 30 d) 1 in 15
- 5) Bitumen grade 80/100 indicates that under the standard test conditions, penetration value of would vary from a) 0.8 mm to 1 mm b) 8 mm to 10 mm c) 8 cm to 10 cm d) 0.08 mm to 0.1 mm
- 6) In cement concrete pavements, tie bars are installed in a) Expansion joints b) Contraction joints c) Warping joints d) Longitudinal joints
- 7) Which of the following transportation modes is most suitable for heavy and bulk goods? a) Railways b) Airways c) Road transport d) Waterways
- 8) The soundness test for aggregate used for, a) hardness b) durability under weather condition c) shape & size d) water absorption
- 9) The corrected modulus of subgrade reaction for standard diameter plate is 6kg/cm³. What would be the modulus of subgrade reaction of the soil when tested with a 30 cm diameter plate ? a) 15kg/cm³ b) 25kg/cm³ c) 30kg/cm³ d) 60kg/cm³
- 10) Speed regulations on roads is decided on the basis of, a) 60th percentile cumulative frequency b) 75th percentile cumulative frequency c) 80th percentile cumulative frequency d) 85th percentile cumulative frequency

- 11) For the construction of Water Bound Macadam (WBM) roads, the correct sequence of operations after spreading coarse aggregates is a) Dry rolling, wet rolling, application of screening and application of filler b) Dry rolling, application of filler, wet rolling and application of screening c) Dry rolling, application of screening, wet rolling, and application of filler d) Dry rolling, application of screening, application of filler and wet rolling
- 12) Which one of these methods of tunnel construction is not suitable in rocks ? a) Full face method b) Compressed air method c) Heading and benching method d) Drift method

Q2. Solve the following.

- A) a) Explain the role of transportation in the development of country and write the different modes of transport. 6
- B) b) Define Stopping Sight Distance (SSD) and calculate the stopping sight distance required on sloping gradient of -2.5% for a design speed of 100 kmph. Assume a reaction time of 2.5 sec, coefficient of friction of 0.7 and brake efficiency of the vehicle is 50%. Also calculate the intermediate sight distance. 6

Q3. Solve the following.

- A) a) What is Traffic Engineering? Explain Traffic characteristics. 6
- B) b) Write a short note on marshal stability test. 6

Q4. Solve Any Two of the following.

- A) a) Write a short note on Highway lighting & Traffic operations 6
- B) b) Enumerate the construction steps of Water Bound Macadam (WBM) road. 6
- C) c) Explain in detail Road accidents & safety. 6

Q5. Solve Any Two of the following.

- A) a) Write short notes on : 1) IRC class AA loading 2) Bridge bearings. 6
- B) b) What are the methods for different types of Pavements. 6
- C) c) Write difference between Airway & Waterway. 6

Q6. Solve Any Two of the following.

- A) a) Discuss the Drainage in pavement design. 6
- B) b) Explain 'Flexible Pavement and Rigid Pavement' and bring out the points of difference. 6
- C) c) Write difference between Railways & Pipeline Transportation. 6

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